OVERVIEW

The dataset contains 551 trading days from Feb 2022 to Mar 2025. Each day, social media users cast Up or Down votes before the market opens, predicting whether HSI will rise or fall. I suggested that votes Up and Down are prediction votes from social media users before the market opens on the same day. That is, users vote on row X before the market opens on row X.

This report is split into two parts:
Part 1 - Data Analysis
Part 2 - Trading Strategy and backtesting

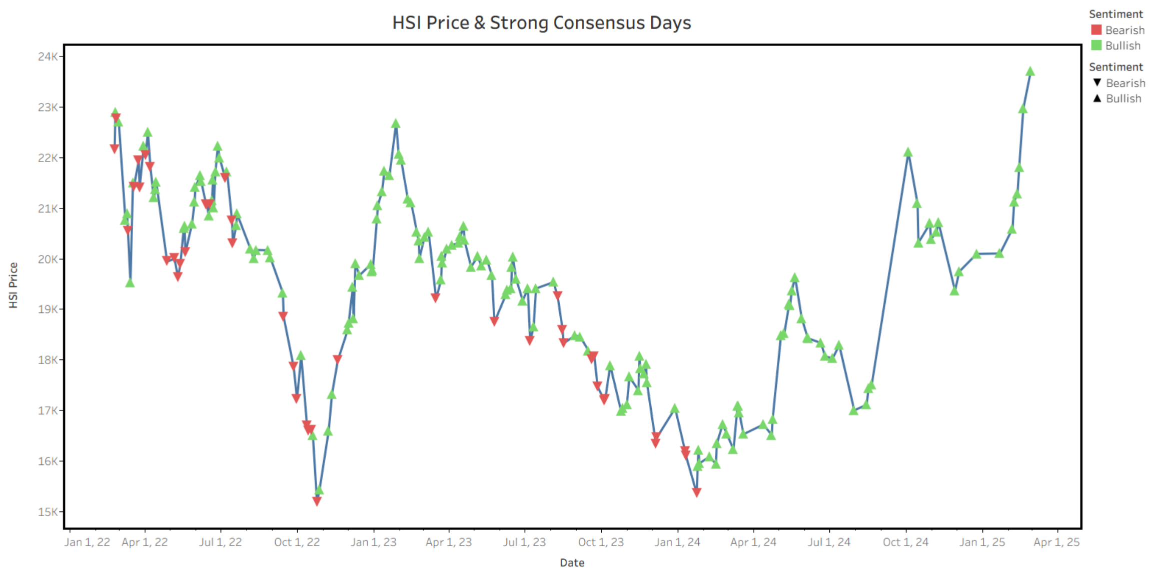
Part 1: Data Analysis

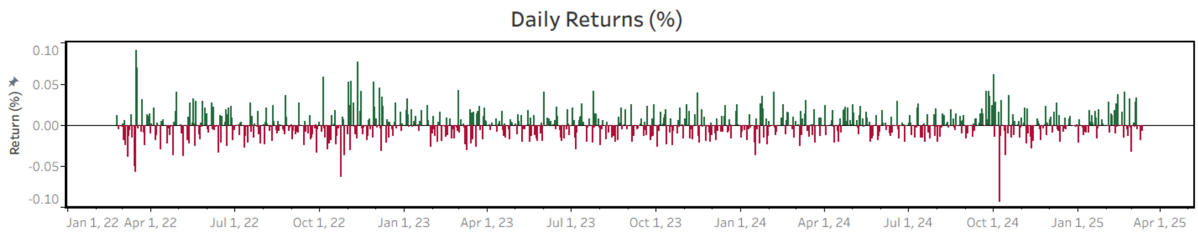
1.1 HSI Price Overview

Over the sample period the HSI ranged from a low of 14,687 to a high of 24,370, with a mean of 19,050. The index was broadly in a downtrend from early 2022 before recovering in 2024. Total length of data - 747. Days with social media votes - 551(excluding first day).

Daily return statistics
Mean daily return: 0.02%
Std of daily return: 1.72% (annualised: 27.32%)
Best single day: +9.08%
Worst single day: -9.4%

The daily return distribution is roughly symmetric around zero





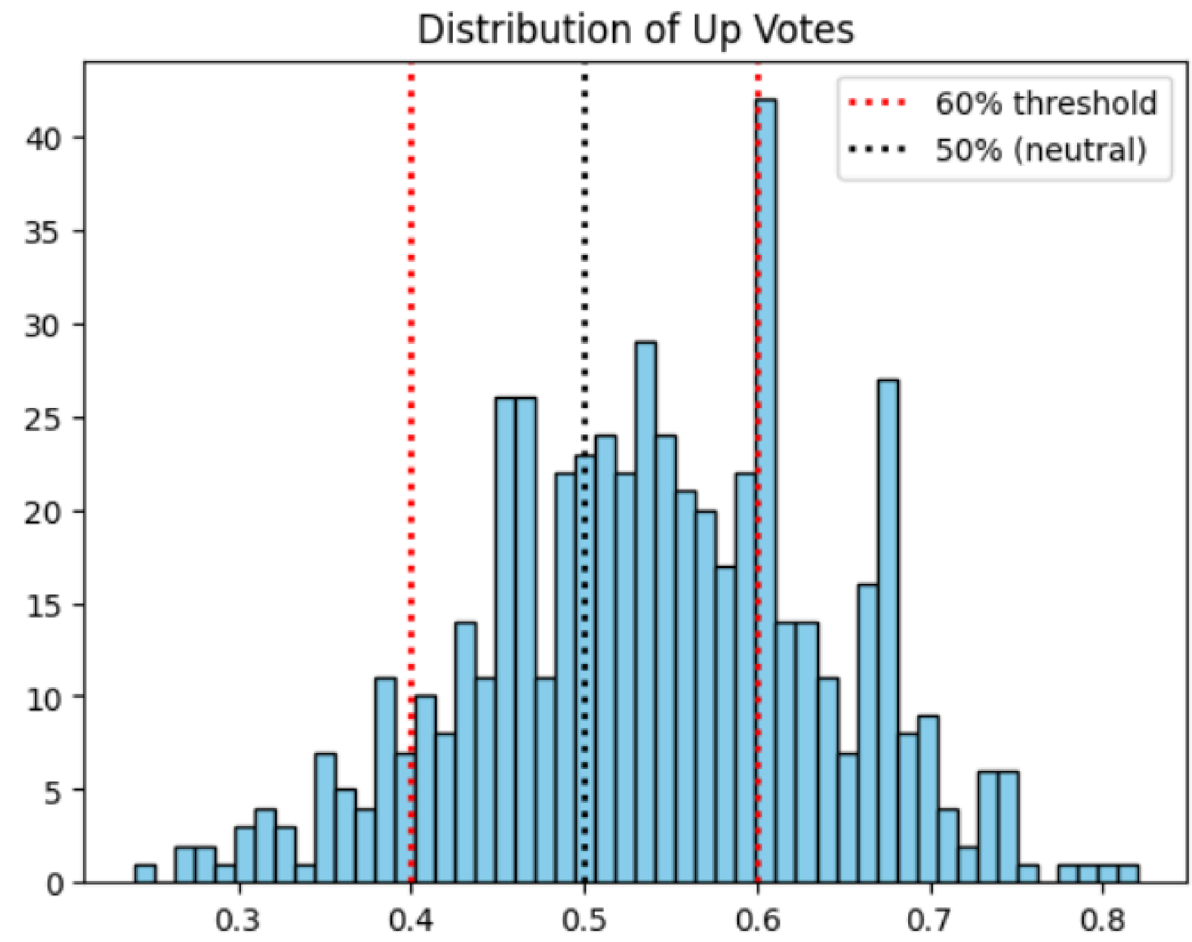
1.2 Vote Distribution Analysis

The Up and Down vote fractions always sum to 1.0 (Up + Down = 1). So analysing Up votes alone is sufficient.

Key observations from the vote data
Mean Up-vote fraction : 0.541 (the crowd is slightly bullish on average)
Std of Up-vote fraction: 0.102
Min / Max Up fraction : 0.24 / 0.82

Breakdown of crowd sentiment
Bullish days (Up > 50%) : 350 days (63.4% of sessions)
Bearish days (Up < 50%) : 179 days (32.4% of sessions)
Strong bull (Up > 60%) : 153 days
Strong bear (Down > 60%): 43 days

The crowd leans bullish overall (up votes average 54%) which likely reflects a general optimism bias among social media users.



1.3 Social Media's ability to predict next day return

Before building a strategy let's find out if there is a correlation between votes and Market Return.

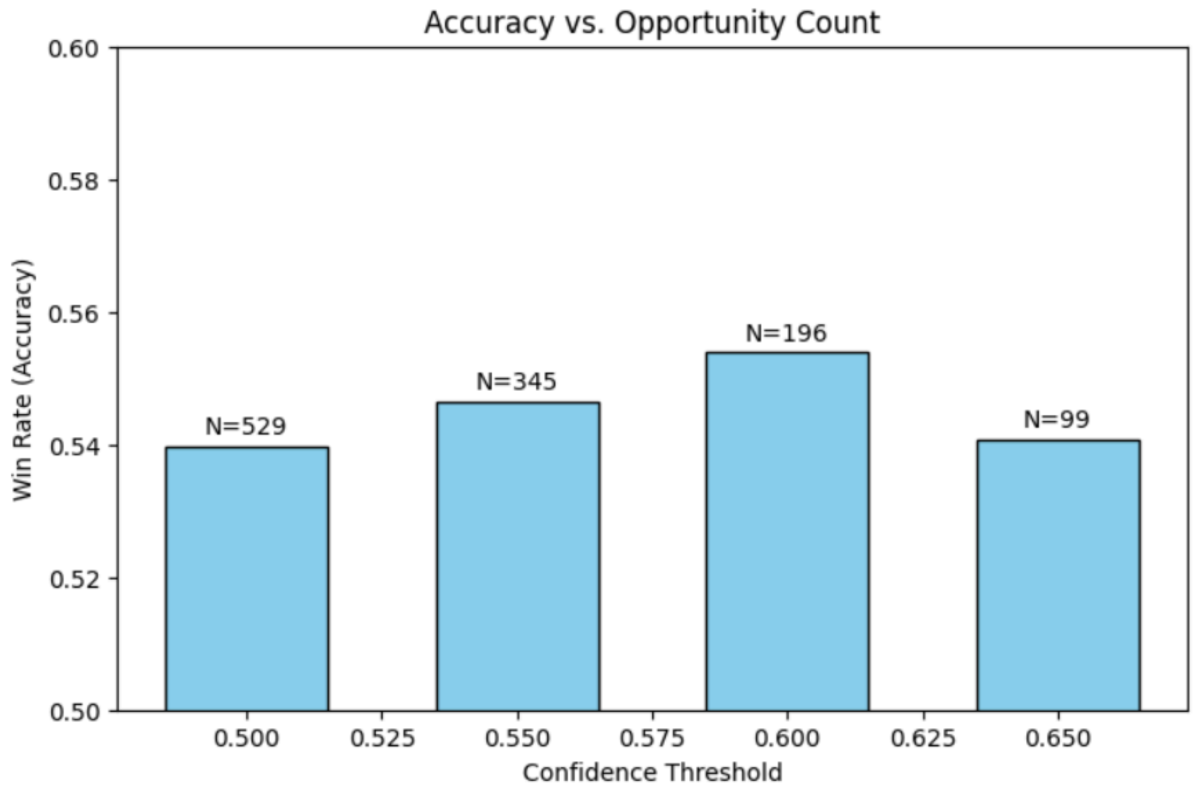
Correlation(Up votes, Market Return) = 0.1713

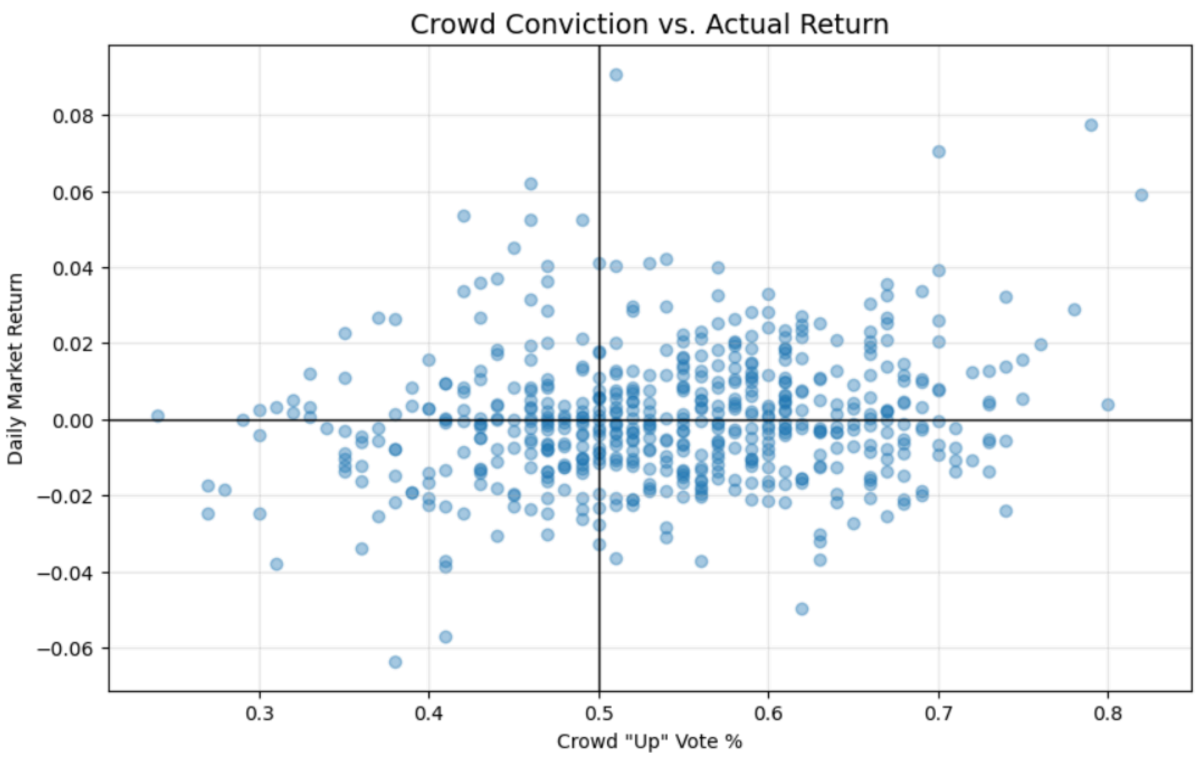
This positive correlation means that there is a weak association between higher up votes and higher market returns. That is, the same as what the crowd expects.

Crowd accuracy by confidence level:
(Accuracy = fraction of days the crowd predicted the direction correctly)

Threshold > 50%: accuracy = 54% (n = 529 days)
Threshold > 55%: accuracy = 54.6% (n = 345 days)
Threshold > 60%: accuracy = 55.4% (n = 196 days)
Threshold > 65%: accuracy = 54.08% (n = 99 days)

As we can see, the crowd is only slightly better than the mere flip of a coin at every confidence level. The improvement from 50% to 60% confidence interval is very small, which then drops at 65%. Correlation analysis showcases us the same story. Crowd adds only a tiny bit of information, but still doesn't show predictive power.





Part 2: Trading Strategy

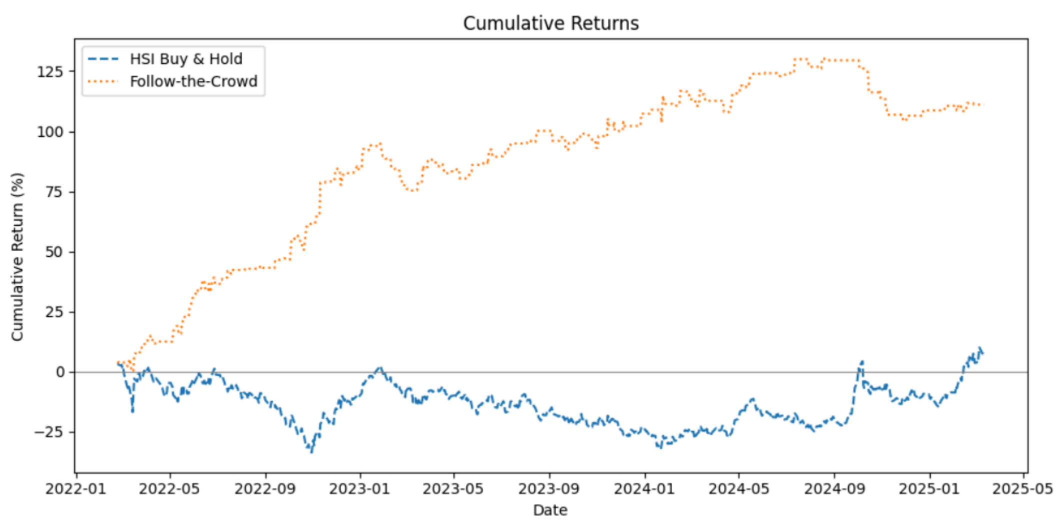
2.1 Strategy Logic

Based on the data analysis in Part 1, the crowd is slightly more correct than wrong. This motivates me to trade only when the crowd shows strong conviction(>60%, since winrate goes down after >60%).

Rules
Long : Up votes > 60% (crowd is bullish, we buy)
Short : Down votes > 60% (crowd is bearish, we sell)
Flat : All other days

Signal breakdown over the full sample
Crowd strongly bullish (Up > 60%) : 153 days -> we go Long
Crowd strongly bearish (Down > 60%): 42 days -> we go Short
No strong consensus : 356 days -> flat

The strategy is selective; it only trades on about 35% of days. On most days it does nothing.



2.2 Conclusion and limitations

Limitations
Small sample: 3 years of data is not enough to be fully confident that the consensus trading strategy is real and not just a coincidence.
Threshold selection: the 60% threshold was chosen after looking at the data, which risks overfitting. It should be tested on new data.
No transaction costs: brokerage fees and bid-ask spreads are not included. These would reduce returns in practice.
Calculation: close-to-close returns are used as a simplification. In practice, the trade would be executed closer to the open.

Conclusion
The analysis suggests that social media sentiment has a weak but positive ability to predict HSI market direction. Crowd accuracy remains only slightly above 50%, meaning sentiment alone is not a highly reliable forecasting tool.

Based on these findings, a trading strategy of strong consensus was chosen that only trades when the crowd sentiment exceeds 60% in either direction. The strategy produced a total return of approximately 111%, significantly outperforming the HSI buy-and-hold return of 6.5% over the sample period.

However, these results should be treated carefully due to several limitations of the analysis.